Data Visualisation and Data-driven Decision Making Project

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Code on GitHub

1 Introduction

We present four visualisations of historic stock market data (Defeat Beta, 2024) from January 2019 to December 2025. These visualisations are ordered in terms of data granularity and are meant to represent different views of the stock market, from subjective personal feelings to objective hard realities. This is done to bridge the mental gap between a trader looking at their portfolio performance and the broader trends of the market, to allow for easier rationalization of their trading decisions. The first visualisation is an interactive exercise showing the disparity between short-term and long-term single-stock performance using line charts. The second visualisation is a heatmap of aggregate industry performance meant to express the waves of gains and losses. The third is a bar chart showcasing quarterly sector performance revealing the broader market trends. And lastly is an animated spiral line chart showing the broadest reality of aggregate market performance.

2 Data Pre-processing

Data is first adjusted for inflation to the 2025 US Dollar. The data is then filtered based on the starting closing price of each stock and the average trading volume, keeping stocks above the 20th percentile, from either its industry, sector, or global group depending on the plot. This is done to prevent penny stocks from skewing the average price and change in percentage, especially as their market importance is highly negligible and only relevant for a small group of people and businesses.

3 Chart Decisions and Justification

When actively engaging in trading it is hard to rationalize the relatively short-term losses while seeing the overall performance of your portfolio, as even if you are up 15% in the year a seemingly sudden 5% loss feels like your choices of trades are wrong, not realizing the possibility of a broader market slowdown. Therefore, we decided to try to visualise this juxtaposition of subjective day-to-day feelings and objective longer-term and broader realities to implore the reader to stay level-headed when faced with short-term challenges whether trading stocks or in everyday life.

3.1 Short-term Long-term Disparity Exercise

With this exercise we use multiple single-stock line charts to try to model the day-to-day experience of trading stocks; looking at lines going up or down, trying to remain optimistic that your choices will result in profit and not loss, but also grounded when short-term spikes occur.

3.2 Daily Industry Performance Heatmap

Showcases the waves of gains and losses across industries, showing the slightly broader and longer term context that is effecting individual stocks. Thus, allowing for more rational thinking about single-stock performance in the industries. To focus on the bigger picture, not the specific details, the industry labels were omitted.

3.3 Quarterly Sector Performance Bars

Shows broad market trends per sector quarter-to-quarter and all time. This level of broadness now shows real market trends with negative quarters likely being indicators of systematic regression in price rather than a momentary slow-down putting more context to industry and single-stock longer-term performance.

3.4 Overall Animated Spiral of Performance

Made as an alternative to a stacked line chart, the spiral allows for comparison of daily aggregate stock trends and performance across years using a continuous line rather than disconnected lines, allowing for animation. It shows us the highs and lows of trading for each year but also across multiple years. This provides full long-term context about the gains or losses of sectors and industries in any period of time as well as serves as point of comparison for the performance of your own portfolio to the total market performance.

References

Defeat Beta (2024). Yahoo finance data dataset. https://huggingface.co/datasets/defeatbeta/yahoo-finance-data. Accessed: 15-04-2026, Licensed under ODC-BY.